

ATPL EASA STUDY GUIDE | 022 - AGK INSTRUMENTATION

Chapter 10: Flight Management Systems (FMS & Cost Index)

Syllabus Source: CAE Oxford (pp. 349-398) | Pass Target: >= 90% AviationExam

1. FMS SYSTEM ARCHITECTURE & DUAL DATABASES

The Flight Management System (FMS) is the brain of modern navigation, integrating lateral routing (LNAV), vertical profile optimization (VNAV), and performance calculations:

- **Hardware Layout:** Consists of two redundant Flight Management Computers (FMCs) cross-talking over ARINC 429/AFDX buses, controlled via pilot Multifunction Control Display Units (MCDU) or Touchscreen Control Units.
- **Navigation Database (NDB):** A certified, READ-ONLY database updated internationally every 28 DAYS according to the ICAO AIRAC cycle. Contains worldwide airports, runways, waypoints, airways, radio aids, SIDs, STARs, and published instrument approaches. The flight crew CANNOT overwrite or alter stored airway points.
- **Performance Database (PDB):** Contains manufacturer aerodynamic drag polars, engine fuel burn curves, weight limits, and flight envelope speeds for the specific airframe/engine type.

2. COST INDEX (CI) & SPEED OPTIMIZATION

The FMS computes the most economical cruising speed and climb/descent profiles based on airline Cost Index (CI):

- **Cost Index Formula:** $\text{Cost Index} = \frac{\text{Cost of Time (\$ per hour)}}{\text{Cost of Fuel (cents per pound or kg)}}$. CI balances flight crew/maintenance hourly operational costs against fuel expenses.
- **Cost Index = 0:** Represents ZERO cost of time! The FMS commands MAXIMUM RANGE CRUISE (MRC) speed: burns minimum possible fuel per nautical mile, resulting in the slowest cruise speed and longest flight duration.
- **Cost Index = 999 (Maximum CI):** Time cost is infinitely more important than fuel! The FMS commands the maximum authorized cruising speed (V_{MO} / M_{MO}), minimizing flight time regardless of high fuel consumption.
- **Long Range Cruise (LRC):** A speed schedule flown at 99% of Maximum Range Cruise efficiency, providing ~3% to 5% higher speed for only 1% extra fuel burn.

Cost Index Value	Aeronautical Optimization Objective	Flight Performance Result
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CI = 0 (Minimum Fuel)	Minimum fuel consumption per nautical mile (Zero time cost).	Maximum Range Cruise (MRC). Lowest cruise speed; maximum flight time; greatest fuel economy.
Intermediate CI (Typical 20-80)	Standard airline operating balance between schedule and fuel burn.	Econ climb, Econ cruise, and Econ descent speeds calculated dynamically.
Maximum CI (e.g. 999)	Minimum flight duration (Time is critical, fuel cost ignored).	Flies at maximum certified speed (V _{MO} / M _{MO}). Highest fuel burn; shortest flight time.

[!] EXAM TRAP (AVIATIONEXAM) - Cost Index Effect on Speeds and Profiles Summary

- INCREASING Cost Index: Increases Climb Speed, Cruise Speed/Mach, and Descent Speed; steepens descent path; shortens flight time! - DECREASING Cost Index: Decreases Climb/Cruise/Descent speeds; flattens descent path; saves fuel! - CI = 0 ALWAYS corresponds to MAXIMUM RANGE (Minimum Fuel)!

3. REQUIRED NAVIGATION PERFORMANCE (RNP) VS ACTUAL (ANP)

Modern RNAV routes are certified under Performance-Based Navigation (PBN) standards:

- Required Navigation Performance (RNP): The lateral accuracy containment standard in nautical miles that the aircraft must maintain for at least 95% of total flight time (e.g. RNP 4 for oceanic, RNP 1 for terminal SIDs/STARs, RNP 0.3 for RNAV/RNP final approaches).
- Actual Navigation Performance (ANP / EPE): The current real-time estimated navigation error computed by the FMS from multi-sensor blending (GPS + DME/DME + IRS). If ANP exceeds RNP: ANP > RNP, an 'UNABLE RNP' caution illuminates on the PFD/ND, requiring an immediate missed approach or ATC notification!

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)**[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): Cost Index Trade-Off**

Formula: Cost Index = Time Cost (\$/hr) / Fuel Cost (cents/kg). - CI = 0: Fuel is everything -> Flies at maximum range speed (MRC) -> MINIMUM FUEL BURN. - CI = Max (99/999): Time is everything -> Flies at VMO/MMO -> MINIMUM FLIGHT TIME regardless of fuel burn.

4. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)**Q1: What is the operational effect of entering a Cost Index (CI) of ZERO into the Flight Management Computer?**

- [A]: The FMC commands maximum certified airspeed
- [B]: The FMC commands Maximum Range Cruise (MRC) speed, resulting in minimum fuel burn per nautical mile
- [C]: The autopilot disconnects
- [D]: The FMS switches to manual navigation

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. When CI = 0, time cost is treated as zero. The computer optimizes purely for fuel economy, flying at Maximum Range Cruise (MRC) speed.

Q2: How often is the worldwide aeronautical Navigation Database (NDB) within an aircraft FMS updated?

- [A]: Every 7 days
- [B]: Every 28 DAYS in accordance with the international ICAO AIRAC cycle
- [C]: Annually on January 1st
- [D]: Whenever the aircraft undergoes a C-check

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. International aeronautical navigation data is published and synchronized on a strict 28-day cycle known as the AIRAC (Aeronautical Information Regulation And Control) cycle.

Q3: What occurs if the Actual Navigation Performance (ANP) computed by the FMS becomes LARGER than the Required Navigation Performance (RNP)?

- [A]: The autopilot increases speed
- [B]: An 'UNABLE RNP' alert is annunciated and the flight crew must execute a missed approach or notify ATC
- [C]: The FMC switches automatically to VOR tracking
- [D]: Fuel is dumped

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. When ANP exceeds RNP, the aircraft can no longer guarantee the required obstacle clearance boundary. Flight crew procedures mandate executing an immediate go-around.